Lista - Inequações do 2º grau, domínio e função - Matemática Elementar

Inequações do 2º grau

13. Resolva as inequações do 2º grau.

1a. $x^2 - x < 0$

$a = 1 \quad b = -1 \quad c = 0$

$\Delta = b^2 - 4ac$

$\Delta = -(-1)^2 - 4.1.0$

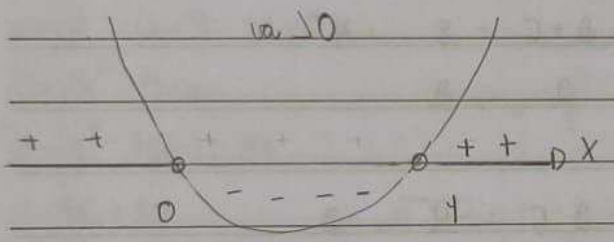
$\Delta = 1 - 0$

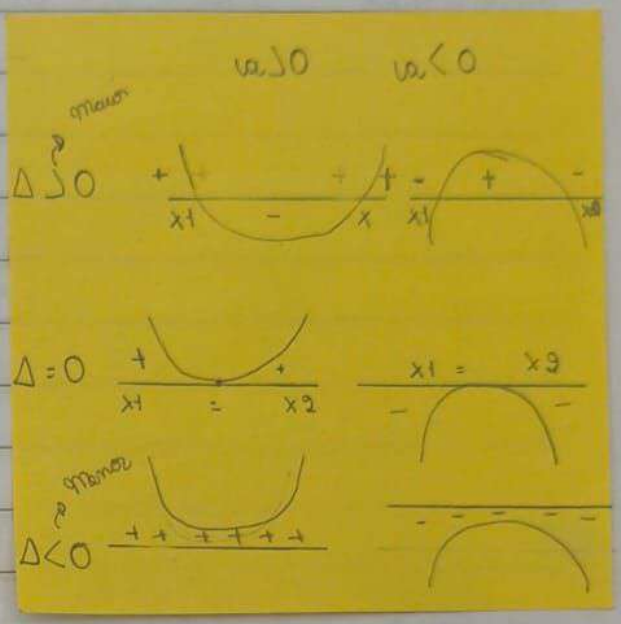
$\Delta = 1$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-(-1) \pm \sqrt{1}}{2.1} = \frac{1 \pm 1}{2}$

$x' = \frac{1+1}{2} = \frac{2}{2} = 1$

$x'' = \frac{1-1}{2} = \frac{0}{2} = 0$





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$f(x) > 0$ quando $x < 0$ ou > 1

$f(x) < 0$ quando $0 < x < 1$

$S =]0, 1[$

b. $4x^2 - 1 \leq 0$

$a = 4 \quad b = 0 \quad c = -1$

$\Delta = b^2 - 4ac$

$\Delta = 0^2 - 4.4(-1)$

$\Delta = 0 + 16$

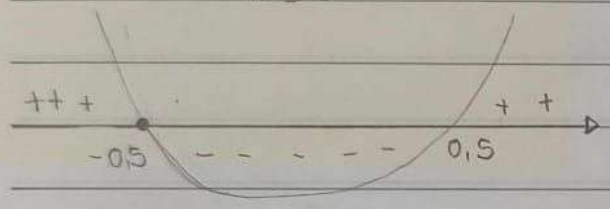
$\Delta = 16$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-0 \pm \sqrt{16}}{2.4} = \frac{0 \pm 4}{8}$

$x' = \frac{0+4}{8} = \frac{4}{8} = 0,5$

$x'' = \frac{0-4}{8} = \frac{-4}{8} = -0,5$

$a > 0$



$f(x) > 0$ quando $x < -0,5$ e $> 0,5$

$f(x) < 0$ quando $-0,5 < x < 0,5$

$S = [-0,5, 0,5]$

c. $x^2 - 2x - 5 > 3$ → maior

$x^2 - 2x - 5 - 3 > 0$

$x^2 - 2x - 8$

$a = 1 \quad b = -2 \quad c = -8$

$\Delta = b^2 - 4ac$

$\Delta = (-2)^2 - 4.1(-8)$

$\Delta = 4 + 32$

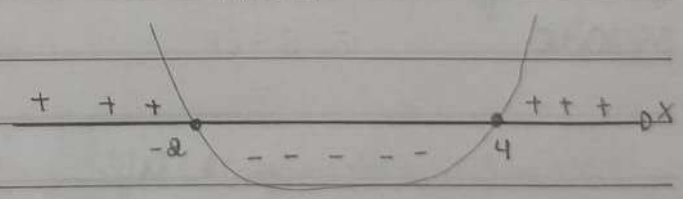
$\Delta = 36$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-(-2) \pm \sqrt{36}}{2.1} = \frac{2 \pm 6}{2}$

$x' = \frac{2+6}{2} = \frac{8}{2} = 4$

$x'' = \frac{2-6}{2} = \frac{-4}{2} = -2$

$a > 0$



$f(x) > 0$ quando $x < -2$ e > 4

$f(x) < 0$ quando $-2 < x < 4$

$S = [-2, 4]$

1d. $4x^2+1>4x$

$4x^2+1-4x>0$

$a=4 \quad b=-4 \quad c=1$

$\Delta=b^2-4ac$

$\Delta=(-4)^2-4.4.1$

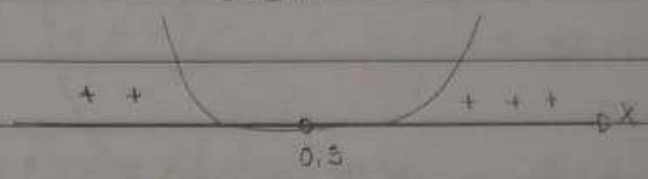
$\Delta=16-16$

$\Delta=0$

$x=\frac{-b\pm\sqrt{\Delta}}{2a}=\frac{-(-4)\pm\sqrt{0}}{2.4}=\frac{4\pm0}{8}$

$x'=\frac{4+0}{8}=\frac{4}{8}=0,5$

$a>0$



$f(x)>0$ quando $x<0,5$ ou $x>0,5$

$S=]0,5[$

1e. $21+4x-x^2>0$

$a=-1 \quad b=4 \quad c=21$

$\Delta=b^2-4ac$

$\Delta=4^2-4(-1).21$

$\Delta=16+84$

$\Delta=100$

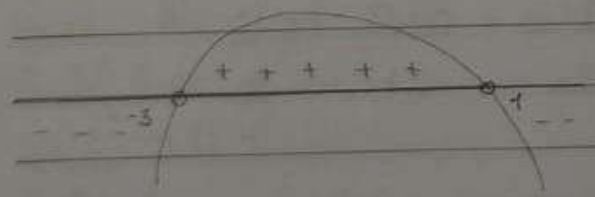
$x=\frac{-b\pm\sqrt{\Delta}}{2a}=\frac{-4\pm\sqrt{100}}{2(-1)}=\frac{-4\pm10}{-2}$

$x'=\frac{-4+10}{-2}=\frac{6}{-2}=-3$

$x''=\frac{-4-10}{-2}=\frac{-14}{-2}=7$

Negativo

$a<0$



$f(x)<0$ quando $x>-3$ ou $x<7$

$f(x)>0$ quando $-3<x<7$

$S=]-3,7[$

if. $x^2 - 2x - 5 > 3$

$x^2 - 2x - 5 - 3 > 0$

$x^2 - 2x - 8 > 0$

$a = 1 \quad b = -2 \quad c = -8$

$\Delta = b^2 - 4ac$

$\Delta = -2^2 - 4 \cdot 1 \cdot -8$

$\Delta = 4 + 32$

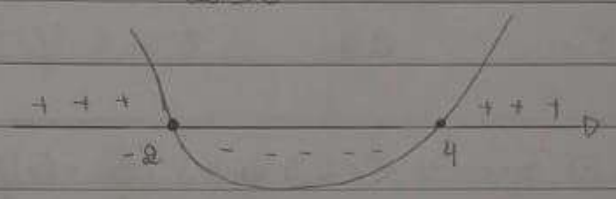
$\Delta = 36$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-(-2) \pm \sqrt{36}}{2 \cdot 1} = \frac{2 \pm 6}{2}$

$x' = \frac{2+6}{2} = \frac{8}{2} = 4$

$x'' = \frac{2-6}{2} = \frac{-4}{2} = -2$

$a > 0$



$f(x) > 0$ quando $x < -2$ ou $x > 4$

$f(x) < 0$ quando $-2 < x < 4$

$S = [-2, 4]$

Domínio das Funções

44. Determine o domínio das seguintes funções:

1a. $f(x) = \frac{1}{x^2 - 4}$

$a = 1 \quad b = 0 \quad c = 4$

$\Delta = b^2 - 4ac$

$\Delta = 0^2 - 4 \cdot 1 \cdot (-4)$

$\Delta = 0^2 + 16$

$\Delta = 16$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-0 \pm \sqrt{16}}{2 \cdot 1} = \frac{0 \pm 4}{2}$

$x' = \frac{0+4}{2} = \frac{4}{2} = 2 \qquad x'' = \frac{0-4}{2} = \frac{-4}{2} = -2$

$D(f) = \{x \in \mathbb{R} / x \neq 2 \text{ ou } x \neq -2\}$

2b. $f(x) = \sqrt{x^2 - 9}$

$a = 1 \quad b = 0 \quad c = 9$

$\Delta = b^2 - 4ac$

$\Delta = 0^2 - 4 \cdot 1 \cdot -9$

$\Delta = 0 + 36$

$\Delta = 36$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-0 \pm \sqrt{36}}{2 \cdot 1} = \frac{0 \pm 6}{2}$

$x' = \frac{0+6}{2} = \frac{6}{2} = 3$

$x'' = \frac{0-6}{2} = \frac{-6}{2} = -3$

$a > 0$

$D(f) = \{x \in \mathbb{R} / x < -3 \text{ ou } x > 3\}$

c. $f(x) = \dfrac{1}{\sqrt{x^2-1}}$

$a=1 \quad b=0 \quad c=-1$

$\Delta = b^2 - 4ac$

$\Delta = 0^2 - 4 \cdot 1 \cdot -1$

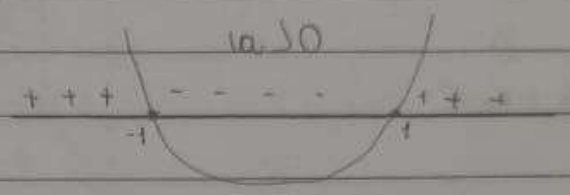
$\Delta = 0 + 4$

$\Delta = 4$

$x = \dfrac{-b \pm \sqrt{\Delta}}{2a} = \dfrac{-0 \pm \sqrt{4}}{2 \cdot 1} = \dfrac{0 \pm 2}{2}$

$x' = \dfrac{0+2}{2} = \dfrac{2}{2} = 1$

$x'' = \dfrac{0-2}{2} = \dfrac{-2}{2} = -1$



$D(f) = \{x \in \mathbb{R} / x < -1 \text{ ou } x > 1\}$

d. $f(x) = \dfrac{1}{1+\sqrt{x}}$

$\sqrt{x} = x > 0$

$1 + x \neq 0$

$\sqrt{x} \neq 1$

$x \neq 1$

$D(f) = \{x \in \mathbb{R} / x > 0, x \neq 1\}$

e. $f(x) = \dfrac{\sqrt{x}}{x^2-9}$

$a=1 \quad b=0 \quad c=-9$

$\Delta = b^2 - 4ac$

$\Delta = 0^2 - 4 \cdot 1 (-9)$

$\Delta = 0 + 36$

$x = \dfrac{-b \pm \sqrt{\Delta}}{2a} = \dfrac{-0 \pm \sqrt{36}}{2 \cdot 1} = \dfrac{-0 \pm 6}{2}$

$x' = \dfrac{-0+6}{2} = \dfrac{6}{2} = 3 \qquad x'' = \dfrac{-0-6}{2} = \dfrac{-6}{2} = -3$

$D(f) = \mathbb{R} \{x \in \mathbb{R} / x \neq 3 \text{ ou } x \neq -3\}$

f. $f(x) = \sqrt{x^2 - 3x - 4}$

$a=1 \quad b=-3 \quad c=-4$

$\Delta = b^2 - 4ac$

$\Delta = -3^2 - 4 \cdot 1 (-4)$

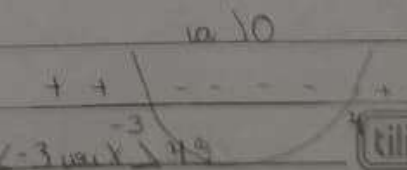
$\Delta = 9 + 16$

$\Delta = 25$

$x = \dfrac{-b \pm \sqrt{\Delta}}{2a} = \dfrac{-(-3) \pm \sqrt{25}}{2 \cdot 1} = \dfrac{3 \pm 5}{2}$

$x' = \dfrac{3+5}{2} = \dfrac{8}{2} = 4$

$x'' = \dfrac{3-5}{2} = \dfrac{-2}{2} = -1$



$D(f) = \{x \in \mathbb{R} / x < -3 \text{ ou } x > 4\}$

g. $f(x) = \frac{1}{\sqrt{2x+3}}$

$2x+3 > 0$

$2x > -3$

$x > \frac{-3}{2}$

$]-\frac{3}{2}, +\infty[$

h. $f(x) = \sqrt{x^2 - x - 6}$

$a = 1 \quad b = -1 \quad c = -6$

$\Delta = b^2 - 4ac$

$\Delta = -1^2 - 4 \cdot 1 \cdot -6$

$\Delta = 1 + 24$

$\Delta = 25$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-(-1) \pm \sqrt{25}}{2 \cdot 1} = \frac{1 \pm 5}{2}$

$x' = \frac{1+5}{2} = \frac{6}{2} = 3$

$x'' = \frac{1-5}{2} = \frac{-4}{2} = -2$

$a > 0$

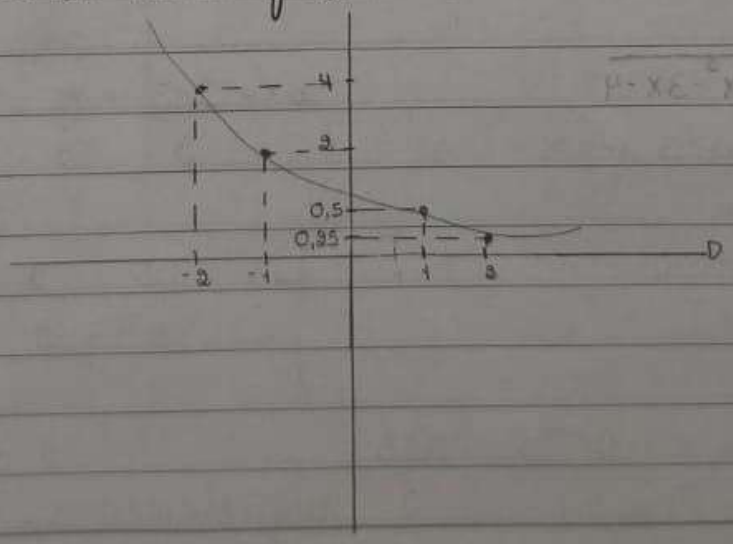
$D(f) = \{x \in \mathbb{R} / x < -2 \text{ ou } x > 3\}$

Função

15. Esboce o gráfico de cada função.

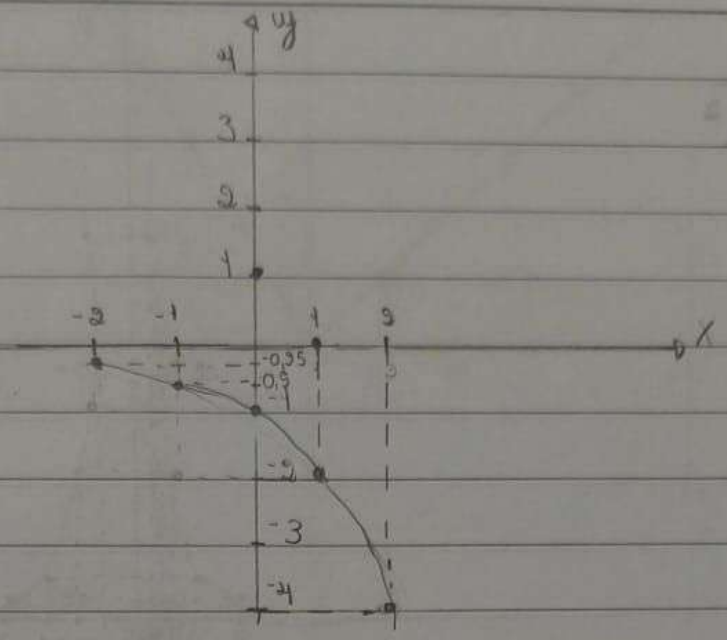
a. $f(x) = 2^{-x}$

x	$f(x)$
1	$2^{-1} = 0,5$
2	$2^{-2} = 0,25$
-1	$2^{-(-1)} = 2$
-2	$2^{-(-2)} = 4$



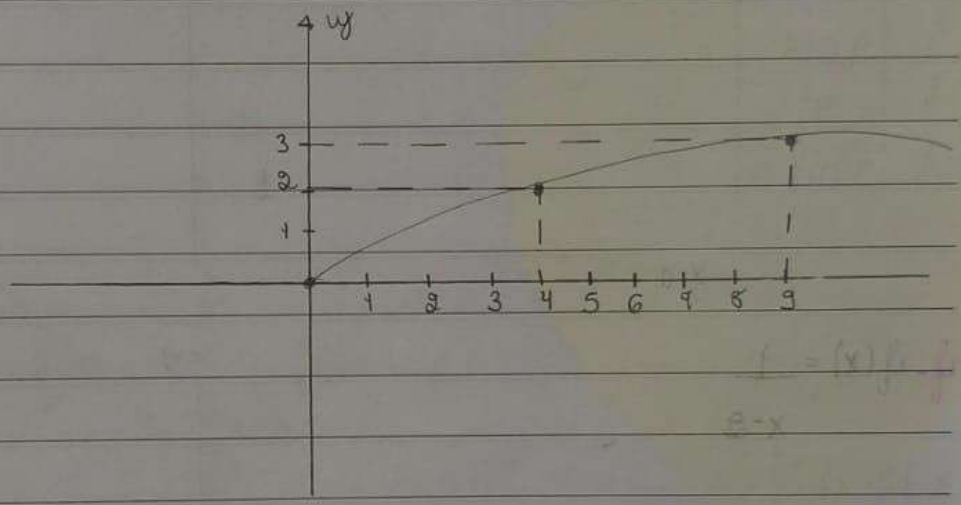
1b. $f(x) = -2^x$

x	f(x)
0	-1
1	-2
2	-4
3	-8
-1	-0,5
-2	-0,25



1c. $f(x) = \sqrt{x}$

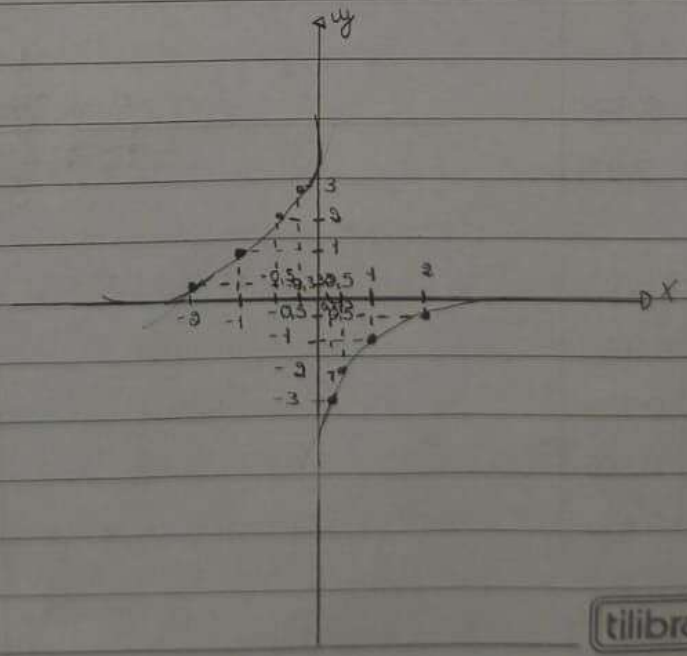
x	f(x)
4	2
9	3



1d. $f(x) = \frac{-1}{x}$

x	f(x)
1	-1
2	-0,5
-1	1
-2	0,5

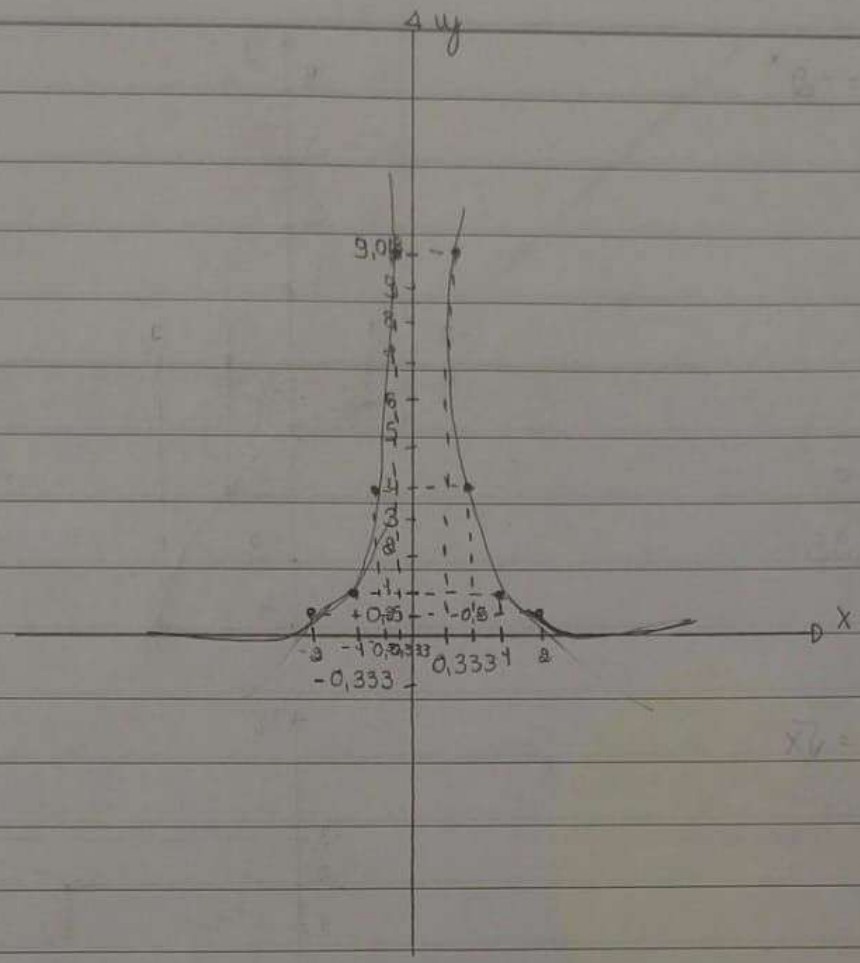
	x	f(x)
0,5	$\frac{1}{2}$	-2
0,333	$\frac{1}{3}$	-3
-0,5	$-\frac{1}{2}$	2
-0,333	$-\frac{1}{3}$	3



1e. $f(x) = \frac{1}{x^2}$

x	f(x)
1	1
2	0,25
-1	1
-2	0,25

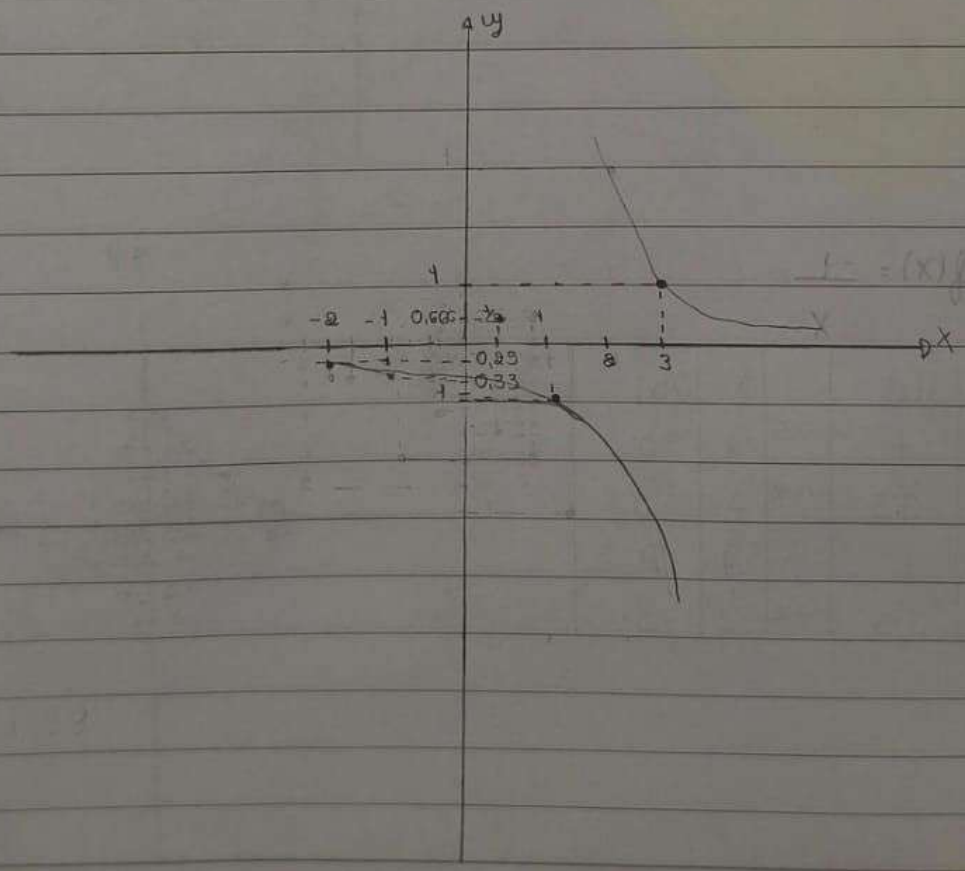
	x	f(x)
0,5	$\frac{1}{2}$	4
0,333	$\frac{1}{3}$	9,018
-0,5	$-\frac{1}{2}$	4
-0,333	$-\frac{1}{3}$	9,018



1f. $f(x) = \frac{1}{x-2}$

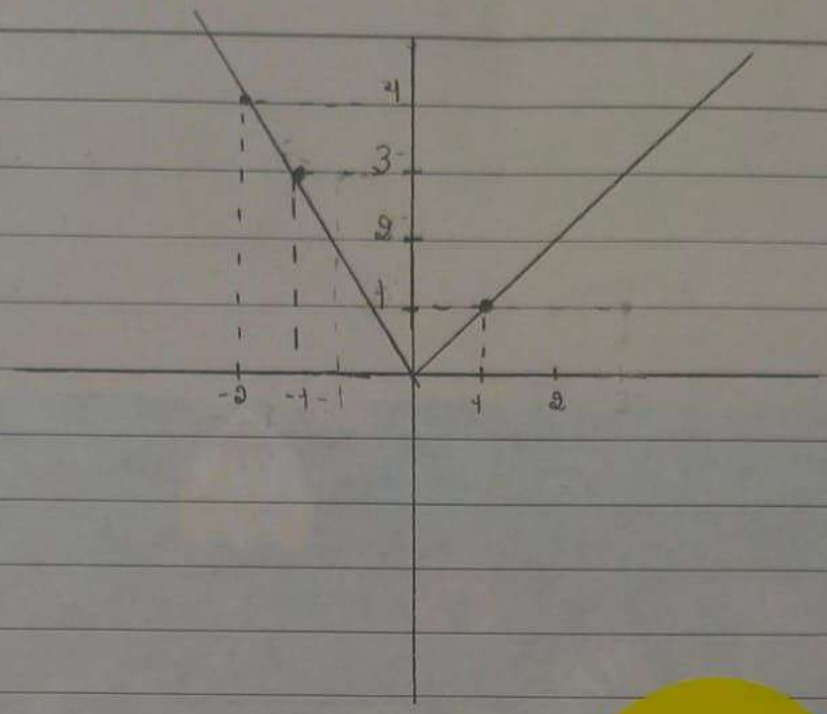
x	f(x)
1	-1
3	1
-1	-0,333
-2	-0,25

	x	f(x)
0,5	$\frac{1}{2}$	0,666
0,333	$\frac{1}{3}$	-0,599
-0,5	$-\frac{1}{2}$	-0,666
-0,333	$-\frac{1}{3}$	-0,498



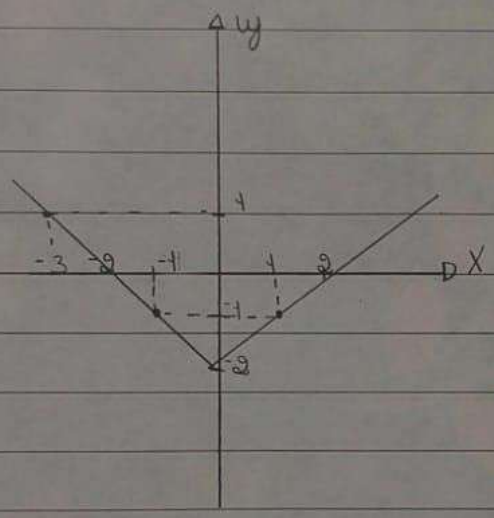
g. $f(x) = |x-2|$

X	f(X)
0	2
1	1
2	0
-1	3
-2	4
-3	5



h. $f(x) = |x| - 2$

X	f(x)
0	0 - 2 = -2
1	1 - 2 = -1
2	2 - 0 = 0
-1	1 - 2 = -1
-2	2 - 2 = 0
-3	3 - 2 = 1



$$|A| = \begin{cases} A, \text{ se } A \geq 0 \\ -A, \text{ se } A < 0 \end{cases}$$